Pengcheng Zhang

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Education

Max Planck Institute for Mathematics, Bonn Jan 2022–Sep 2026
Doctoral student, supervised by Don Zagier

St John's College, University of Oxford, Oxford Oct 2017–Jun 2021
Master of Mathematics in Mathematics

Publications and Preprints

[1] Pengcheng Zhang. *The complexity of Ford domains of* $\Gamma_0(N)$. Results Math., 81(4):Paper No. 100, 2026.

[2] Tian Wang and Pengcheng Zhang. *Ordinary primes for* GL_2*-type abelian varieties and weight* 2 *modular forms*. Mathematika, 72(2):Paper No. e70081, 2026.

[3] Pengcheng Zhang. *On Converse Theorems for Hilbert Modular Forms Assuming Unramified Twists*. Q. J. Math., 76(4):1283–1313, 2025.

[4] Pengcheng Zhang. *The 2-complexity of even positive integers*. Integers, 24:Paper No. A109, 11, 2024.

[5] Benjamin Bedert, George Cooper, Thomas Oliver, and Pengcheng Zhang. *Twisting moduli for* GL(2). J. Number Theory, 217:142–162, 2020.

[6] Pengcheng Zhang. *Elliptic curves and Fourier coefficients of meromorphic modular forms*. preprint on arXiv.

[7] Pengcheng Zhang. *A note on the hypergeometric datum* $\left(\left(\frac{1}{2},\frac{1}{6},\frac{5}{6}\right),(1,1)\right)$ *and symmetric squares of elliptic curves*. exposition.

Research Visit

- Radboud University, Nijmegen, hosted by Wadim Zudilin 17–21 Nov, 2025
- Leiden University, Leiden, hosted by Jan Vonk 28–31 Oct, 2025
- University of Luxembourg, Esch-sur-Alzette, hosted by Michael Daas 14–17 Oct, 2025
- Louisiana State University, Baton Rouge, hosted by Ling Long 29 Mar–4 Apr, 2025
- The Abdus Salam International Centre for Theoretical Physics, Trieste, hosted by Vasily Golyshev and Don Zagier 30 Jun–8 Jul, 2024
- The University of Hong Kong, Hong Kong, hosted by Ben Kane 19–23 Feb, 2024
- SUSTech International Center for Mathematics, Shenzhen, hosted by Don Zagier 22 Nov–17 Dec, 2021

Talks in research seminars/conferences

- *A Habiro formalism for CM points* in Participative Learning Seminar on Any Number Theory (PLeaSANT) at Max Planck Institute for Mathematics, Bonn 13 Apr, 2026
- *Meromorphic modular forms and p-adic Chowla–Selberg periods* in London Junior Number Theory Seminar at King's College, London 31 Mar, 2026

- *Meromorphic modular forms and symmetric powers of elliptic curves* in Number Theory Seminar at University of Luxembourg, Esch-sur-Alzette 14 Oct, 2025
- *Elliptic curves and meromorphic modular forms* in International Workshop on Automorphic Forms at SwissMAP Research Station, Les Diablerets 4 Sep, 2025
- *Coefficients of modular forms modulo (powers of) primes* in Participative Learning Seminar on Any Number Theory (PLeaSANT) at Max Planck Institute for Mathematics, Bonn 26 May, 2025
- *Elliptic curves and coefficients of meromorphic modular forms* in Oberseminar Zahlentheorie at University of Cologne, Cologne 14 Apr, 2025
- *Ordinary primes of weight* 2 *modular forms* in the South Regional Number Theory Conference 2025 at Louisiana State University, Baton Rouge 29 Mar, 2025
- *Elliptic curves and coefficients of meromorphic modular forms* in Oberseminar Anlytic Number Theory and Automorphic Forms at University of Bonn, Bonn 21 Nov, 2024
- *Converse theorems for Hilbert modular forms* in the 36th Automorphic Forms Workshop at Oklahoma State University, Stillwater 23 May, 2024
- *A converse theorem for Hilbert modular forms* in Oberseminar Anlytic Number Theory and Automorphic Forms at University of Bonn, Bonn 9 Nov, 2023
- *Converse theorems for automorphic forms* in St John's College Mathematics Seminar (with Benjamin Bedert) at St John's College, University of Oxford, Oxford 5 Mar, 2020

Talks in learning seminars/study groups

- *Hecke operators on p-adic modular forms* in the IMPRS Seminar on modular forms at Max Planck Institute for Mathematics, Bonn 19 Nov, 2024
- *Orthogonal groups and their symmetric spaces* in the Study Group on Rigid Meromorphic Cocycles for Orthogonal Groups at Max Planck Institute for Mathematics, Bonn 24 Oct, 2024
- Several talks in Exponential Sum Learning Seminar at Max Planck Institute for Mathematics, Bonn 17 Nov, 15 Dec, 2023, and 12 Jan, 2024
- *L-functions and/in Selberg Class* in the IMPRS Seminar on L-functions at Max Planck Institute for Mathematics, Bonn 24 Jan, 2023

Conferences, Workshops, Research Schools Attended

- **Building Bridges 7** at Będlewo Palace 17–21 Aug, 2026
- **Curves, Abelian VArieties and RElated Topics 2** at Universitat de Barcelona, Barcelona 13–17 Jul, 2026
- **p-Adic Kudla program and p-adic automorphic forms** at CIRM, Marseille 8–19 Jun, 2026
- **Abelian varieties and their moduli** at Universitat de Barcelona, Barcelona 3–6 Feb, 2026
- **Luxembourg Number Theory Day 2025** at University of Luxembourg, Esch-sur-Alzette 19 Dec, 2025
- **International Workshop on Automorphic Forms** at SwissMAP Research Station, Les Diablerets 31 Aug–5 Sep, 2025
- **Conference on Progress and Emergent Theories in Zeta and L-Functions** at Max Planck Institute for Mathematics, Bonn 21–25 Jul, 2025
- **33èmes Journées Arithmétiques** at University of Luxembourg, Esch-sur-Alzette 30 Jun–4 Jul, 2025
- **Asymptotic Counting and L-Functions** at Max Planck Institute for Mathematics, Bonn 5–9 May, 2025
- **Southern Regional Number Theory Conference 2025** at Louisiana State University, Baton Rouge 29–30 Mar, 2025
- **The Legacy of John Tate, and Beyond** at Harvard University, Cambridge 17–21 Mar, 2025
- **Arithmetic Geometry in Shenzhen** at Southern University of Science and Technology, Shenzhen 16–20 Dec, 2024
- **New Advances in the Langlands Program: Geometry and Arithmetic** at University of Oxford, Oxford 30 Sep–4 Oct, 2024
- **School on Hodge Theory and Shimura Varieties** at University of Duisburg-Essen, Essen 23–27 Sep, 2024

- **Conference on Arithmetic Geometry** at Max Planck Institute for Mathematics, Bonn 22–26 Jul, 2024
- **Modular Forms, L-functions, and Eigenvarieties** at École Normale Supérieure, Paris 18–21 Jun, 2024
- **The 36th Automorphic Forms Workshop** at Oklahoma State University, Stillwater 20–24 May, 2024
- **Shanks Conferences on L-functions and Automorphic Forms** at Vanderbilt University, Nashville 13–16 May, 2024
- **Periods in Physics, Number Theory and Algebraic Geometry** at Hausdorff Research Institute for Mathematics, Bonn 22–26 Apr, 2024
- **International Conference on Modular Forms and q-Series** at University of Cologne, Cologne 11–15 Mar, 2024
- **Modular Curves and their Arithmetic** at University of Warwick, Coventry 6–8 Dec, 2023
- **MM(P): Mahler Measures of Polynomials** at Radboud University, Nijmegen 24–27 Oct, 2023
- **Modular Forms and Arithmetic Geometry** at TU Darmstadt, Darmstadt 18–22 Sep, 2023
- **Young Number Theorists in Bonn** at Max Planck Institute for Mathematics, Bonn 11–15 Sep, 2023
- **Arithmetic of automorphic forms** at Laboratoire de Mathématiques d'Orsay, Orsay 4–8 Sep, 2023
- **Early Number Theory Researchers Workshop 2023** at University of Bielefeld, Bielefeld 23–25 Aug, 2023
- **Summer School on the Arithmetic of the Langlands Program** at University of Bonn, Bonn 8–12 May, 2023
- **Arithmetic Statistics, Automorphic Forms, and Ergodic Methods** at Max Planck Institute for Mathematics, Bonn 24–28 Apr, 2023
- **Special values of L-functions** at Paderborn University, Paderborn 13–17 Mar, 2023
- **Conference on Motives and Automorphic Forms** at Max Planck Institute for Mathematics, Bonn 6–10 Mar, 2023
- **Number theory meets p-adic representations** at University of Münster, Münster 13–17 Feb, 2023
- **Luxembourg Number Theory Day 2022** at University of Luxembourg, Esch-sur-Alzette 16 Dec, 2022
- **Early Number Theory Researchers Workshop 2022** at TU Darmstadt, Darmstadt 26–28 Oct, 2022

Other experiences

- Referee for Results in Mathematics, Proceedings of the American Mathematical Society, and Research in Number Theory
- Senior Counselor at Ross Mathematics Program (Ross/Asia) in Huangshan 9 Jul–10 Aug, 2018
- Junior Counselor at Ross Mathematics Program (Ross/Asia) in Huangshan 10 Jul–11 Aug, 2017

Languages

Chinese, English